Change of Interval & Functions having Arbitrary Period

Let the function $f(x)$ be defined in the interval ~~$(0, 2c)$~~ $(-c, c)$. Now, we change the function to the period 2π.

$\because$ $2c$ is the interval for the variable x

$\because$ 1 is the interval for the variable $\frac{x}{2c}$

$\because$ 2π " " " " " " $\frac{2\pi x}{2c} = \frac{\pi x}{c}$

Let, $z = \frac{\pi x}{c} \Rightarrow x = \frac{cz}{\pi}$

Now, the function $f(x)$ of period $2c$ is transformed to $F(z) = f\left(\frac{cz}{\pi}\right)$ of period 2π.

$\therefore$ The fourier series is,

$$F(z) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nz + \sum_{n=1}^{\infty} b_n \sin nz$$

where, $a_0 = \frac{1}{\pi}\int_0^{2\pi} F(z)\, dz$

$$a_n = \frac{1}{\pi}\int_0^{2\pi} F(z)\cos nz\, dz$$

$$b_n = \frac{1}{\pi}\int_0^{2\pi} F(z)\sin nz\, dz$$

Putting $z = \frac{\pi x}{c}$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}\left(a_n \cos\frac{n\pi x}{c} + b_n \sin\frac{n\pi x}{c}\right)$$

Here,

$$a_0 = \frac{1}{\pi}\int_0^{2\pi} f\left(\frac{cz}{\pi}\right) dz$$

$$\Rightarrow a_0 = \frac{1}{\pi}\int_0^{2c} f(x)\, d\left(\frac{\pi x}{c}\right)$$

$$\Rightarrow a_0 = \frac{1}{c}\int_0^{2c} f(x)\, dx$$

Also,

$$a_n = \frac{1}{\pi}\int_0^{2\pi} f\left(\frac{cz}{\pi}\right)\cos nz\, dz$$

$$\Rightarrow a_n = \frac{1}{\pi}\int_0^{2c} f(x)\cos\left(\frac{n\pi x}{c}\right) d\left(\frac{\pi x}{c}\right)$$

$$\Rightarrow a_n = \frac{1}{c}\int_0^{2c} f(x)\cos\left(\frac{n\pi x}{c}\right) dx$$

Similarly,

$$b_n = \frac{1}{c}\int_0^{2c} f(x)\sin\left(\frac{n\pi x}{c}\right) dx$$

Questions

Q.1) Find the fourier series for the following function,

$$f(x) = \begin{cases} x, & 0 < x < 1 \\ 1-x, & 1 < x < 2 \end{cases}$$

Sol^n:

The fourier series is,

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}\left(a_n \cos\left(\frac{n\pi x}{c}\right) + b_n \sin\left(\frac{n\pi x}{c}\right)\right)$$

Now,

$$a_0 = \frac{1}{c}\int_0^{2c} f(x)\, dx$$

$$= \frac{1}{1}\left[\int_0^1 x\, dx + \int_1^2 (1-x)\, dx\right.$$

$$= \left[\frac{x^2}{2}\right]_0^1 + \left[x - \frac{x^2}{2}\right]_1^2$$

$$a_0 = \frac{1}{2} - 0 + 2 - \frac{2^2}{2} - 1 + \frac{1}{2}$$

$$\Rightarrow a_0 = 0$$

Then,

$$a_n = \frac{1}{c}\int_0^{2c} f(x)\cos\left(\frac{n\pi x}{c}\right) dx$$

$$= \frac{1}{1}\left[\int_0^1 x.\cos\left(\frac{n\pi x}{1}\right) dx + \int_1^2 (1-x).\cos\left(\frac{n\pi x}{1}\right) dx\right]$$

$$= \left[x\frac{\sin(n\pi x)}{n\pi} - (1)\left(\frac{-\cos(n\pi x)}{n^2\pi^2}\right)\right]_0^1$$

$$+ \left[(1-x)\frac{\sin(n\pi x)}{n\pi} - (-1)\left(\frac{-\cos(n\pi x)}{n^2\pi^2}\right)\right]_1^2$$

$$= \left[x\frac{\sin(n\pi x)}{n\pi} + \frac{\cos(n\pi x)}{n^2\pi^2}\right]_0^1$$

$$+ \left[(1-x)\frac{\sin(n\pi x)}{n\pi} - \frac{\cos(n\pi x)}{n^2\pi^2}\right]_1^2$$

$$= \left[\cancelto{0}{\frac{\sin n\pi}{n\pi}} + \frac{\cos n\pi}{n^2\pi^2} - 0 - \frac{\cos 0}{n^2\pi^2}\right]$$

$$+ \left[-\frac{\sin(2n\pi)}{n\pi} - \frac{\cos(2n\pi)}{n^2\pi^2} - 0 + \frac{\cos n\pi}{n^2\pi^2}\right]$$

$$a_n = 0 + \frac{(-1)^n}{n^2\pi^2} - \frac{1}{n^2\pi^2} + 0 - \frac{1}{n^2\pi^2} - 0 + \frac{(-1)^n}{n^2\pi^2}$$

$$\Rightarrow a_n = \frac{2}{n^2\pi^2}\left[(-1)^n - 1\right]$$

Also,

$$b_n = \frac{1}{c}\int_0^{2c} f(x)\sin\left(\frac{n\pi x}{c}\right)dx$$

$$= \frac{1}{1}\left[\int_0^1 x\sin\left(\frac{n\pi x}{1}\right)dx + \int_1^2 (1-x)\sin\left(\frac{n\pi x}{1}\right)dx\right]$$

$$= \left[x\left(\frac{-\cos(n\pi x)}{n\pi}\right) - (1)\left(\frac{-\sin(n\pi x)}{n^2\pi^2}\right)\right]_0^1$$

$$+ \left[(1-x)\left(\frac{-\cos(n\pi x)}{n\pi}\right) - (-1)\left(\frac{-\sin(n\pi x)}{n^2\pi^2}\right)\right]_1^2$$

$$= \left[\frac{-\cos n\pi}{n\pi} + \frac{\sin n\pi}{n^2\pi^2} + \frac{0.\cos 0}{n\pi} - \frac{\sin 0}{n^2\pi^2}\right]$$

$$+ \left[\frac{\cos 2n\pi}{n\pi} - \frac{\sin 2n\pi}{n^2\pi^2} + 0 + \frac{\sin n\pi}{n^2\pi^2}\right]$$

$$= -\frac{(-1)^n}{n\pi} + 0 + 0 - 0 + \frac{1}{n\pi} - 0 + 0 + 0$$

$$\Rightarrow b_n = \frac{1}{n\pi}\left[-(-1)^n + 1\right]$$

Hence, the fourier series is,

$$f(x) = 0 + \sum_{n=1}^{\infty}\left[\frac{2}{n^2\pi^2}\left[(-1)^n - 1\right]\cos(n\pi x)\right]$$

$$+ \sum_{n=1}^{\infty}\left[\frac{1}{n\pi}\left[-(-1)^n + 1\right]\sin(n\pi x)\right]$$

$$\Rightarrow f(x) = -\frac{4}{\pi^2}\left[\frac{\cos \pi x}{1^2} + \frac{\cos 3\pi x}{3^2} + \dots\right]$$

$$+ \frac{2}{\pi}\left[\frac{\sin \pi x}{1} + \frac{\sin 3\pi x}{3} + \dots\right]$$

Ans